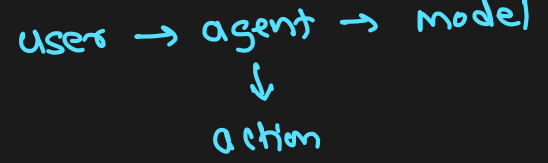
Human In the Loop

What is HITL?

- Human-in-the-Loop (HITL) is a design pattern in AI systems where human judgment is integrated into the automated decision-making process at critical points
- Instead of allowing AI to operate completely autonomously, HITL creates a collaboration where:
  - AI proposes actions, recommendations, or decisions
  - Humans review, validate, or modify these proposals
  - Only then are actions executed
- AI has fundamental limitations
  - **Hallucinations:**
    - AI confidently generates false information
    - It cannot distinguish between fact and fiction
  - **Lack of True Understanding**
    - AI processes patterns, not meaning
    - It doesn't truly "understand" context, nuance, or consequences
  - **Bias Amplification**
    - AI inherits biases from training data
    - It amplifies historical inequalities



The Problem with Agentic AI

- Agentic AI doesn't just answer questions—it takes actions too

- Delete database records
  - IRREVERSIBLE
  - Data loss, business disruption
- Send emails
  - IRREVERSIBLE
  - Reputation damage, legal liability
- Process payments
  - IRREVERSIBLE
  - Financial loss, fraud
- Update production code
  - IRREVERSIBLE
  - System crashes, security breaches
- Modify legal documents
  - IRREVERSIBLE
  - Contract violations, lawsuits
- Schedule appointments
  - Partially reversible
  - Wasted time, customer dissatisfaction

With or Without HITL

- **Without HITL**



- **With HITL**



Why is it required?



- **AI is not reliable enough** to act without oversight
- **AI is not accountable** for its actions
- **AI lacks context** that only humans possess
- **AI cannot evolve** with changing intent
- **Laws demand** human oversight
- **Trust requires** human involvement
- **The cost of failure** is too high

The Design Pattern

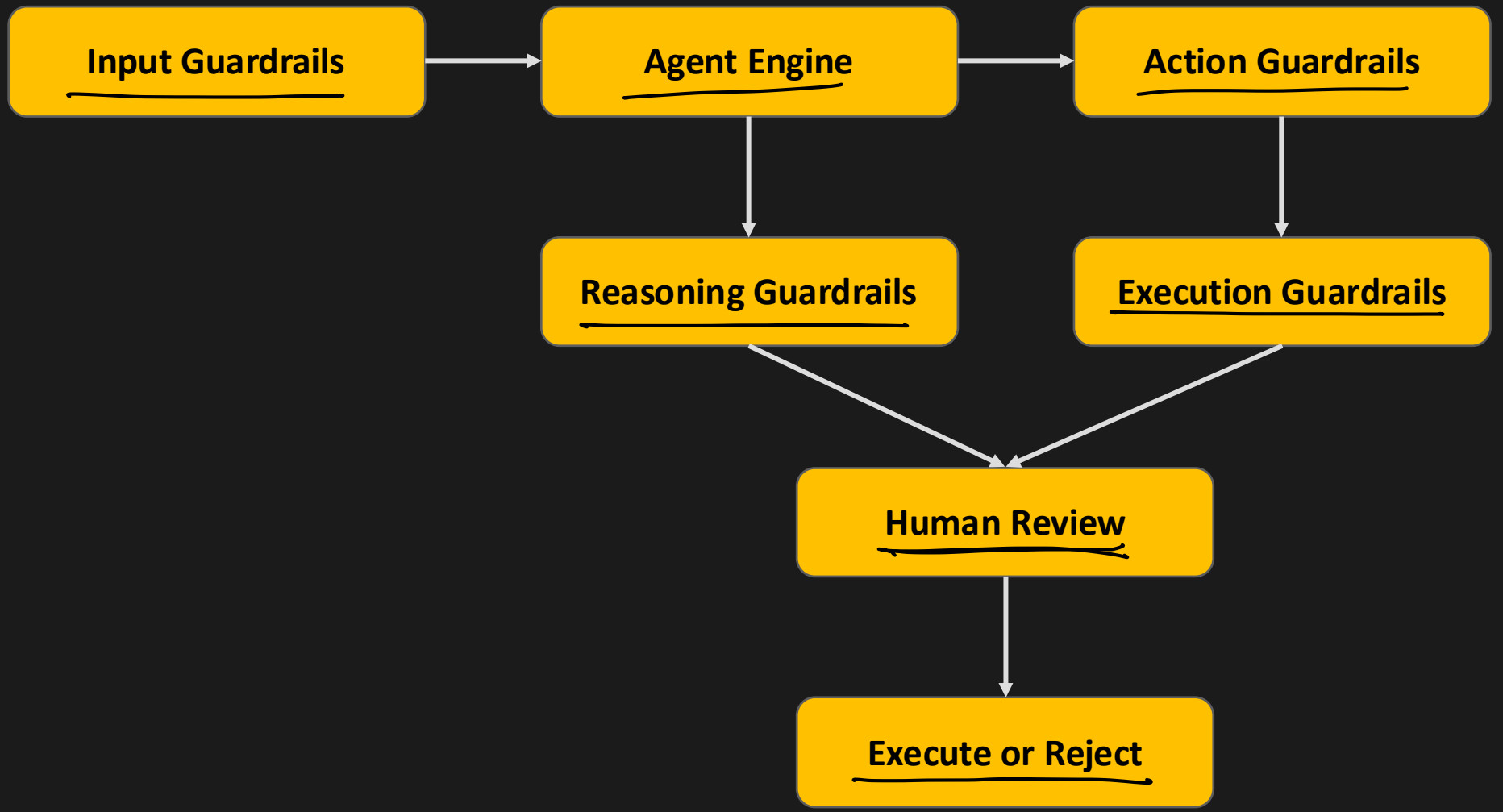
- HITL is not a single "approval button." It exists on a spectrum of human involvement, from high control to high automation
- Modern architectures have evolved to implement these patterns effectively

Pattern	Agent Autonomy	Human Role	Best For
Human-in-Command	Low	Drives every step	High-risk Legal/Medical decisions
Human-as-Filter	Medium	Approves/Edits final output	Content Generation (e.g., marketing materials)
Human-as-Backup	High	Only intervenes on error/uncertainty	Customer Support, most production workflows
Human-on-the-Loop	Max	Audits logs after completion	High-volume analysis with low risk

Specific Implementation Patterns

- **Approval based**
  - Agent proposes action → Human approves or rejects
  - Binary decision point
  - Use case: Loan approvals, compliance checks, high-stakes binary choices
- **Review and Edit**
  - Humans actively modify AI-generated content
  - Iterative refinement process
  - Use case: Report drafting, marketing copy, creative content
- **Escalation based**
  - System dynamically hands over to human during interaction
  - Triggered by: User request, negative sentiment, low confidence
  - Use case: Customer service, complex problem-solving
- **Confirm and then execute**
  - Agent prepares action → User confirms → Execution
  - Simple, low-risk pattern
  - Use case: Scheduling meetings, sending emails
- **Propose and then review**
  - Action sent to internal reviewer for sign-off
  - Multiple approval levels possible
  - Use case: Refunds, financial adjustments, change requests

Architectural Components



Guardrails

- **Input Guardrails** (PII)
  - Check user requests for prompt injection
  - Detect out-of-scope content
  - Filter inappropriate language
  - Validate input format
- **Reasoning Guardrails**
  - Control how agent decides what to do
  - Prevent unsafe tool combinations
  - Enforce reasoning constraints
  - Block disallowed reasoning paths
- **Output Guardrails**
  - Check final responses for hallucinations
  - Verify policy compliance
  - Detect sensitive data leakage
  - Validate output format
- **Action Guardrails**
  - Check what agent is about to do
  - Validate action against policies
  - Require confirmation for high-risk actions
  - Log all actions for audit

State Management & Interrupts

- **Interrupts and breakpoints**
  - Hardcoded pause points before sensitive tools
  - Agent state is "frozen" at breakpoint
  - Human sends approve/reject signal
  - Execution resumes only after signal received
  - **Example:** delete_user() tool triggers breakpoint
- **State Injection**
  - Reviewer can "go back" to previous step
  - Edit the agent's state (observations, thoughts)
  - Resume execution from edited state
  - Allows precise steering without starting over
- **Confidence based Escalation**
  - Agent calculates uncertainty score
  - If score exceeds threshold → Pause
  - Escalate to human for decision
  - Works with: Probability scores, confidence intervals